A Theory of Economic Slack

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Draft version: July 2026
Draft URL: pascalmichaillat.org/18/

CHAPTER 3.

Matching function

In slackish markets, all trades are mediated by a matching function. These markets therefore differ from Walrasian markets, where all trades are mediated via an auction, and have sharply different properties.

In this chapter we introduce the matching function, which is the key tool we use to model slackish markets and study slack and unemployment. The matching function summarizes the complex process through which buyers find sellers: for instance, how workers searching for jobs meet firms searching for employees, and how firms searching for customers meet consumers searching for goods and services.[1] Pissarides (2000, pp. 3–4) explains the role of the matching function wonderfully in the case of the labor market, but this book's argument is that his description in fact applies to the vast majority of markets. Slightly generalizing his words:

> Trade in [any] market is a nontrivial economic activity because of the existence of heterogeneities . . . and information imperfections. If all [buyers and sellers] were identical to each other, and if there was perfect information about their [attributes], trade would be trivial. But without homogeneity on either side of the market and with costly acquisition of information, [buyers and sellers] find it necessary to spend resources to find [desirable] trades. . . . The matching function gives the outcome of the investment of resources by [buyers and sellers] in the trading process as a function of inputs. It is a modeling device that captures the implications of the costly trading process without the need

[1] See Petrongolo and Pissarides (2001, pp. 425–427) for a detailed history of the matching function.

to make . . . the features that give rise to it explicit.

After having introduced a generic matching function and discussed its properties, we will cover several specific matching functions.

3.1. Why do we need a matching function?

At any point in time, in almost any market, we have buyers and sellers who want to trade, but are not able to trade immediately. In the labor market, there are workers who want to sell their labor and firms who want to buy labor and yet, even though these two groups coexist, it takes time for workers to find a job and for firms to fill a vacant job. On the housing market, families trying to sell their houses coexist with families looking for a house to buy—but it takes time for families to sell their homes and for other families to find a new home. On the car market, there are cars that sit in inventory and people who are looking to buy cars. It takes time for people to find a car, and it also takes time for the cars in inventory to be sold. On the service market, there may be babysitters who are looking for work, and at the same time, there are families looking to hire—it still takes time for them to find the right babysitter. There are restaurant tables that are empty and hungry people looking for a good restaurant to eat at.

The bottom line is that in almost all markets, buyers who want to buy and sellers who want to sell always coexist—it takes time for the transaction to occur. To model such markets, we cannot use a Walrasian market. In the Walrasian market, one can always buy and sell any quantity of a good at the market price, immediately and without constraints. So a Walrasian market ignores the fact that it takes time and effort to buy and sell in most real-world markets.

To model this complex trading process, we introduce a tool called the matching function. The matching function is an aggregate function that captures and summarizes, at the macro level, all the complexities of trading that happen at the micro level. It is very similar to the production function, which summarizes at the aggregate level all the production that happens at the micro level. It is a simple, well-behaved function that depends on only a few aggregate variables.

Most trading is quite complicated—it is hard for buyers and sellers to get together to execute a trade that both sides desire. For instance, in the labor market, a worker with specific skills would look for a job with specific requirements, whether it be a specific industry, specific location, or specific working conditions or responsibilities. Similarly, a firm advertising a vacant job would have specific needs and would be looking for a worker with the right skill set, the right experience, the right qualifications and character.

This complexity is not limited to the labor market. It would also occur when firms trade with other firms. When firms look for suppliers for a specific part in a product that they are building or for a specific tool, they have a long list of technical requirements because that part is unique to their product or the tool is unique to their manufacturing process. They also have additional constraints on the quality of the part or tool, its reliability, the supplier's location and how it operates. The production capacity of the supplier and its existing commitments also matter, as they determine whether the supplier can produce the part or tool rapidly.

And the complexity of trading is not confined to the labor market or to sophisticated goods and services traded by firms. Even for seemingly simple goods and services purchased in everyday life, trading is often complex because buyers have heterogeneous needs and sellers offer differentiated products. Consider something as simple as a haircut. Hairdressers specialize: for women, men, children; for budget or high-end styles. Hairdressers have more or less experience, are more or less skilled, and are more or less talkative. Hair salons have different numbers of chairs and existing customers so they are more or less busy. And customers' preferences differ. Hours of operation and locations may or may not fit a customer's schedule and daily routine. As a result, it takes time and effort for customers to obtain a suitable haircut, and conversely, for hairdressers to attract and retain customers.

This complexity exists in most markets, so the matching function can be applied to almost all markets in which trading is complex. The only exceptions are auction markets, where the goods that are traded are available in large quantity and are homogeneous. Then, as all goods are the same, finding the right seller is not relevant. Goods traded on such auction markets include some securities—especially stocks—and some commodities—silver, gold, wheat, or cotton (Okun 1981, p. 134). These markets are well represented by a Walrasian model; in fact they are exactly the sort of markets that inspired Walras (Walker 1990, p. 653). But very few markets operate that way; in practice, most goods and services offered for sale are different, require shopping by buyers, and take time to sell for sellers. Even in the financial world that inspired Walras, many markets are better modeled using a matching function than a Walrasian apparatus: these include over-the-counter markets for securities—including corporate and government bonds, derivatives, and some stocks—but also credit markets.[2]

3.2. Definition of the matching function and market tightness

The matching function is an aggregate function that summarizes all the micro-level complexities of the trading process. Consider a market for a specific good with $s > 0$ sellers and $b > 0$ buyers, open for one period. Each seller places one good for sale, and each buyer aims to purchase one good. The matching function $m(s, b)$ gives the number of trades in

[2] See Wasmer and Weil (2004) and (Hugonnier, Lester, and Weill 2025).

the period.

Before going further, we introduce a new, fundamental variable: the market tightness, defined by

$$\theta = \frac{b}{s}.$$

On any market, the tightness is the ratio between the number of buyers and the number of sellers—the two inputs into the matching function.

Market tightness plays a crucial role in the book because it summarizes the state of any market. This is because, under the assumptions that we make, the trading probabilities given by the matching function only depend on market tightness.

Finally, we introduce the matching elasticity to characterize the shape of the matching function. The matching elasticity is the elasticity of the matching function with respect to the number of sellers:

$$\eta = \frac{s}{m(s,b)} \cdot \frac{\partial m(s,b)}{\partial s}. \tag{3.1}$$

The matching elasticity gives the percentage change in the number of trades when the number of sellers increases by 1%. Under our assumptions, the matching elasticity is a function of market tightness, $\eta(\theta)$.

3.3. Assumptions about the matching function

We make a few assumptions about the matching function, to reflect natural properties of markets and ensure that the resulting market model is well behaved. Researchers often assume specific functional forms for their matching functions—typically a Cobb-Douglas function. But most models continue to behave in a similar fashion as long as the matching function satisfies the assumptions listed here. Throughout the book, we stick to these assumptions.

First, the matching function is assumed to be twice continuously differentiable. This ensures that the matching function is smooth enough.

Second, naturally, the matching function is assumed to be strictly positive.

Third, the matching function is assumed to be zero without buyers or sellers: $m(s,b) \to 0$ when $s \to 0$ or $b \to 0$. This is also natural, as people are needed on both sides of the market for trades to occur.

Fourth, the matching function is assumed to be strictly increasing in each argument, s and b. This means that if there are more sellers in the market, or more buyers in the market, there are more trades. This is natural too: if more goods are available, chances are that a greater number of buyers find an appropriate good to buy; conversely, if there are more buyers on the market, chances are that a greater number of sellers find a buyer for their good.

Fifth, the matching function is assumed to have constant returns to scale:

$$m(zs, zb) = zm(s, b) \tag{3.2}$$

for any $z \geq 0$. This assumption critically simplifies the analysis, because it guarantees that the trading probabilities can be expressed as functions of market tightness. Because of constant returns to scale, the elasticity of the matching function with respect to the number of buyers is directly connected to the matching elasticity:

$$\frac{b}{m(s, b)} \cdot \frac{\partial m(s, b)}{\partial b} = 1 - \eta. \tag{3.3}$$

This result is a direct application of Euler's theorem for homogeneous functions (result A.2).

Sixth, the matching function is assumed to be strictly concave in at least one argument, s or b. Because of constant returns to scale, this assumption implies three other properties of the matching function (appendix D). First, the matching function is strictly concave in each argument ($\partial^2 m/\partial s^2 < 0$ and $\partial^2 m/\partial b^2 < 0$). Second, sellers and buyers are Edgeworth-Pareto complements in the matching function ($\partial^2 m/\partial b \partial s > 0$). Third, the matching function is jointly concave in both arguments. However, the joint concavity is weak and not strict because the matching function's Hessian is rank one. The sixth assumption implies that the matching function exhibits diminishing marginal returns to the numbers of sellers and to the number of buyers, which seems natural.

Seventh, the matching elasticity $\eta(\theta)$ is assumed to be increasing in tightness θ. The assumption captures the natural idea that as the market becomes tighter, and sellers are in smaller number relative to buyers, additional sellers contribute proportionally more to total trades.

A final, eighth assumption is that the number of trades that occur within the period considered is less than the numbers of sellers and buyers: $m(s, b) \leq \min(s, b)$. Indeed, there cannot be more goods sold than sellers, and more goods bought than buyers. This assumption is specific to discrete-time models: in a continuous-time model, the matching function gives the flow of trades occurring at any point in time, so there is no restriction on the level of $m(s, b)$.

3.4. Trading probabilities

The matching function $m(s, b)$ tells us that, during a certain period, not all s sellers are able to sell their good, and not all b buyers are able to buy a good. Therefore, we need to figure out the probability that a seller is able to sell,

$$f = \frac{m(s, b)}{s},$$

and the probability that a buyer is able to buy,

$$q = \frac{m(s, b)}{b}.$$

Our assumptions about the matching function have clear implications for how these trading probabilities behave.

3.4.1. Selling probability

We first look at the selling probability:

$$f = \frac{m(s, b)}{s} = m\left(\frac{s}{s}, \frac{b}{s}\right) = m\left(1, \frac{b}{s}\right),$$

since the matching function has constant returns to scale. We can re-express our selling probability as a function of market tightness:

$$f(\theta) = m(1, \theta). \tag{3.4}$$

Since the matching function is strictly increasing in its second argument, we infer that the selling probability is strictly increasing in tightness. The tighter the market, the more likely you are to sell. This makes sense because the tighter a market is, the more buyers there are for each seller. And since the matching function is strictly concave in its second argument, we infer that the selling probability is strictly concave in tightness. This means that a higher tightness leads to a higher selling probability, but with diminishing returns.

We can also see that when tightness is 0, the selling probability must be 0. This is because, in the matching function, when any of the two arguments is 0, there is no trade. This is intuitive because if there are no buyers, there is no chance of selling anything, so the probability of selling is 0.

Lastly, we know that the selling probability is always between 0 and 1 because the matching function is always less than the minimum of its two arguments. Since $m(s, b) \leq \min(s, b)$, then $m(s, b) \leq s$ and $f(\theta) \leq 1$.

3.4.2. Buying probability

Now, we shift our attention to the buying probability:

$$q = \frac{m(s,b)}{b} = m\left(\frac{s}{b}, \frac{b}{b}\right) = m\left(\frac{s}{b}, 1\right),$$

since the matching function has constant returns to scale. Using market tightness, we can rewrite the probability as:

$$q(\theta) = m\left(\frac{1}{\theta}, 1\right), \tag{3.5}$$

so that the buying probability only depends on tightness.

We can go over the properties of the buying probability in the same way. Because the matching function is strictly increasing in its first argument, the buying probability is strictly decreasing in tightness. This means that a buyer in a tight market is less likely to be able to buy the good they want, since there are very few sellers and a lot of buyers, increasing competition. This is true in any tight market: it is a good time to be a seller but a bad time to be a buyer.

Moreover, at the limit where tightness is infinite, the buying probability is 0. The reason is that at the limit, $1/\theta \to 0$, and $m(0, b) = 0$. This makes sense because when infinitely many buyers are competing to buy a good they want, the probability of buying that good is bound to go to 0.

Finally, we know that the buying probability is always between 0 and 1 because the matching function is always less than the minimum of its two arguments. Since $m(s, b) \leq \min(s, b)$, then $m(s, b) \leq b$ and $q(\theta) \leq 1$.

3.4.3. Relation between the trading probabilities

There is a simple relationship between the two trading probabilities. The number of trades is $m(s, b) = f(\theta)s = q(\theta)b$, so $f(\theta)/q(\theta) = b/s = \theta$. That is, for any matching function:

$$f(\theta) = \theta q(\theta). \tag{3.6}$$

From this result we also see that the elasticities of the trading probabilities with respect to tightness are necessarily related:

$$\epsilon_\theta^f = 1 + \epsilon_\theta^q.$$

From the expression (3.5) and result B.7, we see that the elasticity of the buying probability is directly related to the matching elasticity:

$$\epsilon^{q}_{\theta} = -\eta. \tag{3.7}$$

Thus, the elasticity of the selling probability is also related to the matching elasticity:

$$\epsilon^{f}_{\theta} = 1 - \eta. \tag{3.8}$$

Once again, these elasticities might or might not be constant, depending on the matching function.

3.5. Urn-ball matching function

We now study the urn-ball matching function. This matching function is useful because it has a simple microfoundation, which draws on the urn-ball model in probability theory.

3.5.1. Foundation

Several researchers proposed an urn-ball foundation to explain why trading probabilities might be less than 1 in a given period.

Butters (1977) used the urn-ball framework in the product market. In this model, firms advertise their products by dropping ads into customers' mailboxes. The probability to sell is less than 1 because several firms might drop their ad in the same mailbox, in which case the customer only buys from the cheapest firm. The probability to buy is also less than 1, because a customer might not receive any ad in their mailbox.

Hall (1979) used a similar setup in the labor market. In this model, firms simultaneously and randomly make job offers to job seekers. The probability to hire a job seeker is less than 1 because several firms might make a job offer to the same job seeker, who cannot accept more than one offer. The probability to find a job is also less than 1 because a job seeker might be unlucky and not receive any of the job offers.

Both of these situations correspond to an urn-ball setup. In Butters's example, the ads are the balls and the mailboxes are the urns. In Hall's example, the job offers are the balls and the job seekers are the urns.

In this section, we consider as an example the haircut market. There are b customers who are looking for a haircut, and s hairdressers. Each hairdresser can only accommodate one customer at a time, so if two customers pick the same hairdresser, only one can get a haircut. Because customers do not know where the others go, they don't know which hairdressers already serve a customer and which hairdressers are idle. Such coordination failure means that some customers cannot get a haircut and some hairdressers do not get customers.

3.5.2. Expression

We now derive the number of haircuts sold in a simple situation: $b > 0$ customers in need of a haircut each go to one of $s > 0$ open hair salons. The customers pick their hair salons simultaneously and randomly. That day, customers can only make one trip to the hair salon.

The probability that a specific customer goes to a specific hairdresser is $1/s$, and the probability that the customer does not go to the hairdresser is $1 - 1/s$. Accordingly, the probability that a specific hairdresser does not get a visit from any of the b customers is $(1 - 1/s)^b$. Conversely, the probability that a specific hairdresser gets at least one visit is $1 - (1 - 1/s)^b$. Since a hairdresser sells a haircut if at least one customer visits, this probability gives the probability to sell a haircut for a specific hairdresser, as well as the expected number of haircuts sold by the hairdresser (since the hairdresser sells either 0 or 1 haircut). Then, the expected number of haircuts sold by all hairdressers is just

$$m(s, b) = s\left[1 - \left(1 - \frac{1}{s}\right)^b\right].$$

This is the expected number of trades that occur on the haircut market. This matching function is a little cumbersome, but it can be greatly simplified when the number of hairdressers is large enough.

To simplify the above matching function, we use a linear approximation of the logarithm: $\ln(1 - x) \approx -x$ when x is small. This approximation allows us to rewrite the probability that a hairdresser gets no visit at all:

$$\left(1 - \frac{1}{s}\right)^b = \exp\left(b \cdot \ln\left(1 - \frac{1}{s}\right)\right) \approx \exp\left(b \cdot \frac{-1}{s}\right) = \exp\left(-\frac{b}{s}\right).$$

We can then rewrite our matching function, which is the expected number of haircuts sold:

$$m(s, b) = s\left[1 - \exp\left(-\frac{b}{s}\right)\right]. \tag{3.9}$$

This is the urn-ball matching function, which gives the number of trades for a given number of sellers, s, and buyers, b.

3.5.3. Properties

By looking at the matching function (3.9), we can verify that it satisfies all the assumptions listed in section 3.3.

First, we confirm that the matching function is smooth, strictly positive, and zero at the limit without sellers or buyers.

We easily see that the matching function has constant returns to scale:

$$m(zs, zb) = zs\left[1 - \exp\left(-\frac{zb}{zs}\right)\right] = zm(s, b).$$

We check that the matching function is strictly increasing in each argument. We immediately see from (3.9) that if the number of buyers goes up, the number of haircuts also increases. Formally, the partial derivative of the matching function with respect to the number of buyers is simply:

$$\frac{\partial m}{\partial b} = \exp\left(-\frac{b}{s}\right) > 0.$$

Showing that the number of haircuts increases with the number of hairdressers is a little bit trickier since s has two opposite influences on the matching function. We must calculate the derivative formally to check that it is positive. After simplifying, we get:

$$\frac{\partial m}{\partial s} = 1 - \left(1 + \frac{b}{s}\right)\exp\left(-\frac{b}{s}\right). \tag{3.10}$$

We know that $1 + x < \exp(x)$ for all $x > 0$ (result A.8). Thus, $(1 + x)\exp(-x) < 1$ for all $x > 0$, so we verify that $\partial m / \partial s > 0$. From this we conclude that the matching function is strictly increasing in s.

Moreover, the matching function is less than its two arguments. Since $\exp(-x) > 0$, it is clear that $m(s, b) \leq s$. Using again $\exp(x) \geq 1 + x$, we obtain

$$m(s, b) \leq s\left[1 - \left(1 - \frac{b}{s}\right)\right] = b.$$

So overall $m(s, b) \leq \min(s, b)$.

We verify that $\partial m / \partial b$ is strictly decreasing in b, which tells us the matching function is strictly concave in the number of buyers.

Next, we compute the matching elasticity of the urn-ball matching function to better understand the shape and behavior of the matching function. In particular, we show that the urn-ball matching function satisfies the assumption that the matching elasticity is increasing in tightness.

We start from the definition of the matching elasticity, given by (3.1). Using the partial derivative (3.10), we can write the matching elasticity as

$$\eta = \frac{s}{m(s, b)} \cdot \left[1 - \exp\left(\frac{-b}{s}\right)\right] - \frac{b}{m(s, b)} \cdot \exp\left(\frac{-b}{s}\right).$$

Given that the matching function $m(s, b)$ is given by (3.9), we express the matching elasticity as a function of the market tightness:

$$\eta(\theta) = 1 - \frac{\theta}{\exp(\theta) - 1}. \tag{3.11}$$

Hence, the matching elasticity is an increasing function of tightness, growing from 0 when tightness is 0, to $1 - 1/(e - 1)$ when tightness is 1, to 1 when tightness is infinite.

The properties of the matching elasticity appear once we write the exponential function using its Taylor series (result A.10), which then gives:

$$\frac{\exp(\theta) - 1}{\theta} = 1 + \sum_{n=1}^{\infty} \frac{\theta^n}{(n+1)!}.$$

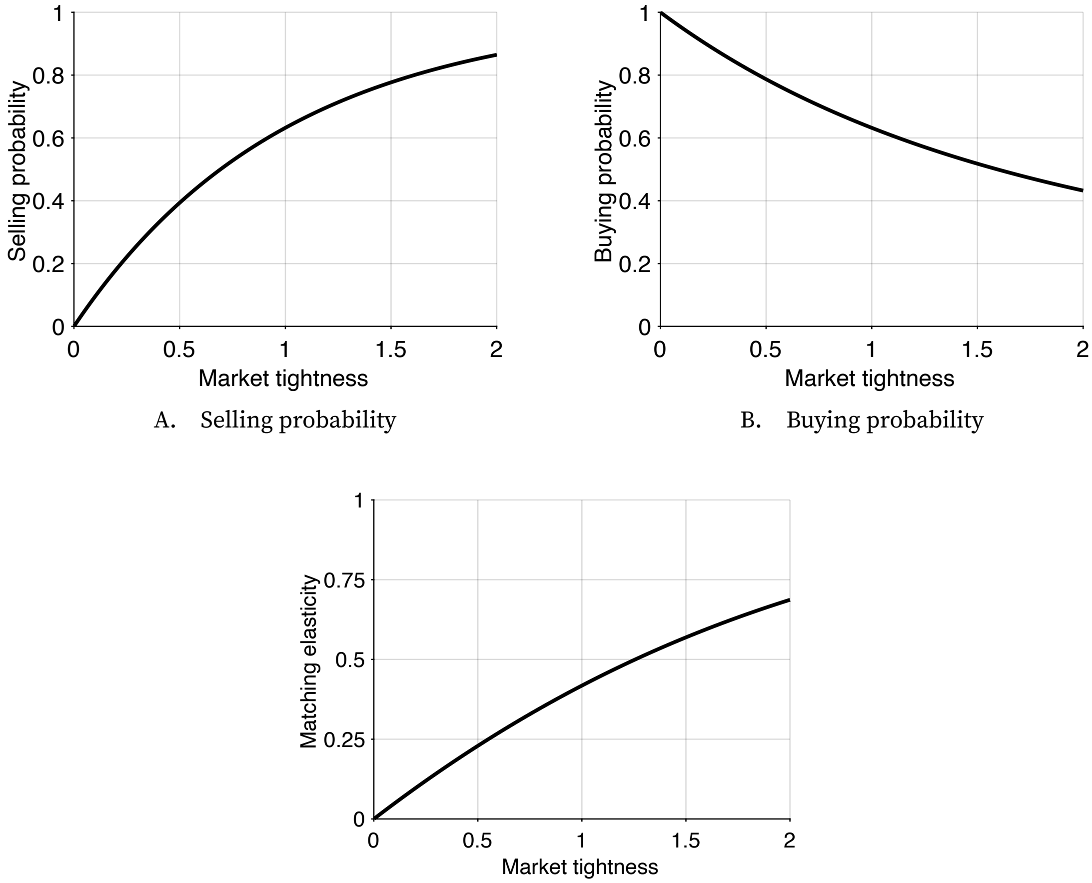
All the terms $\theta^n/(n+1)!$ in the sum are increasing in $\theta \geq 0$, so the function $[\exp(\theta) - 1]/\theta$ is increasing in θ, which tells us that $\eta(\theta)$ is increasing in θ. All the terms in the sum are 0 when $\theta = 0$, which tells us that $\eta(0) = 0$. And all the terms in the sum grow to ∞ when $\theta \to \infty$, which tells us that $\eta(\theta) \to 1$ as $\theta \to \infty$.

The behavior of the elasticity means that when there are no buyers on the market ($\theta = 0$), adding more sellers does not lead to any more trades. The intuition is that the market is already extremely congested with sellers, so more sellers do not bring more trade: only more buyers would generate more trades. On the other hand, when there are almost no sellers ($\theta \to \infty$), adding 1% more sellers generates 1% more trades, even with a fixed number of buyers. In that case the number of trades is almost solely determined by the number of sellers, so they grow proportionally.

Lastly, the selling and buying probabilities given by the urn-ball matching function and its matching elasticity are illustrated in figure 3.1.

3.6. Constant-elasticity-of-substitution matching function

Despite its simple microfoundation, the urn-ball matching function is not used often in theoretical work, because it is a little cumbersome. Instead, it is often preferable to use the constant-elasticity-of-substitution (CES) matching function, as it is much more tractable. While it does not have a microfoundation as simple as the urn-ball process, it is possible to obtain a CES matching function from a queuing process in which sellers call buyers to advertise their goods (Stevens 2007), and from a network in which sellers and buyers are connected (Angelis and Bramoulle 2026).



A. Selling probability

B. Buying probability

C. Matching elasticity

FIGURE 3.1. Properties of the urn-ball matching function

The urn-ball matching function is given by (3.9). The selling and buying probabilities are computed from the matching function with (3.4) and (3.5). The matching elasticity is given by (3.11).

3.6.1. Expression

The CES matching function takes the following form:

$$m(s, b) = \left(s^{-\sigma} + b^{-\sigma}\right)^{-1/\sigma}. \tag{3.12}$$

The parameter $\sigma > 0$ governs the elasticity of substitution between s and b.

3.6.2. Properties

We again check that the assumptions on the matching function listed in section 3.3 are satisfied here.

First, the CES matching function is smooth, strictly positive, strictly increasing in each argument, and zero at the limit without buyers or sellers.

The CES function has obviously constant returns to scale:

$$m(zs, zb) = \left[(zs)^{-\sigma} + (zb)^{-\sigma}\right]^{-1/\sigma} = zm(s, b).$$

Next, we verify that the CES matching function is less than the minimum of its two arguments. Since $b^{-\sigma} > 0$, then $s^{-\sigma} + b^{-\sigma} > s^{-\sigma}$. Hence,

$$m(s, b) = \left(s^{-\sigma} + b^{-\sigma}\right)^{-1/\sigma} < \left(s^{-\sigma}\right)^{-1/\sigma} = s.$$

Similarly, we find $m(s, b) < b$, which overall tells us that $m(s, b) \leq \min(s, b)$.

To verify that the function is concave in the number of sellers, we compute its derivatives. The partial derivative with respect to s is

$$\frac{\partial m}{\partial s} = \left[1 + \left(\frac{s}{b}\right)^{\sigma}\right]^{-\frac{1+\sigma}{\sigma}}.$$

The partial derivative is not only positive but also strictly decreasing in s. From this we infer that the CES matching function is strictly concave in the number of sellers.

Next, we verify that the CES matching elasticity is increasing in tightness. We compute the matching elasticity (3.1) using the results from appendix B:

$$\eta(\theta) = \frac{-1}{\sigma} \cdot \frac{s^{-\sigma}}{s^{-\sigma} + b^{-\sigma}} \cdot (-\sigma)$$

so that

$$\eta(\theta) = \frac{1}{1 + \theta^{-\sigma}}. \tag{3.13}$$

We confirm that the matching elasticity is a strictly increasing function of tightness, growing from 0 when tightness is 0, to 1/2 when tightness is 1, to 1 when tightness is infinite. Just like in the case of the urn-ball matching function, the behavior of the matching elasticity implies that when there are no buyers on the market, adding more sellers does not lead to any more trades, while when there are almost no sellers, numbers of sellers and trades grow proportionally.

With the CES matching function, it is easy to see what happens when there are very few sellers and buyers. We can rewrite the matching function (3.12) as

$$m(s, b) = b \cdot \left[1 + \theta^{\sigma}\right]^{-1/\sigma},$$

where $\theta = b/s$ is the market tightness. When tightness is close to 0, θ^{σ} is negligible compared to 1, so the matching function approximates the number of buyers: $m(s, b) \approx b$. This means that when the number of buyers is small, the number of buyers gives the number

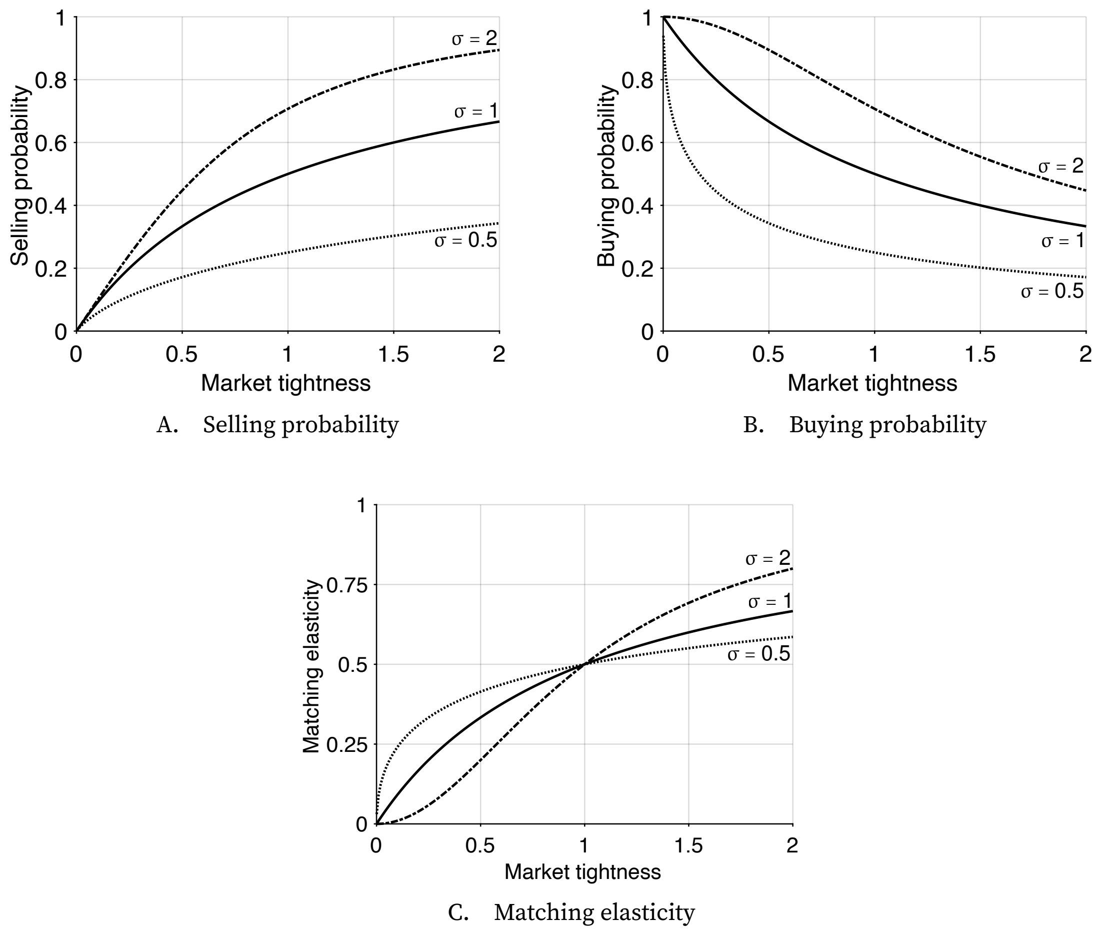


FIGURE 3.2. Properties of the CES matching function

The CES matching functions are given by (3.12) with $\sigma = 0.5$, $\sigma = 1$, and $\sigma = 2$. The selling and buying probabilities are computed from the matching functions with (3.4) and (3.5). The matching elasticity is given by (3.13).

of trades, irrespective of the number of sellers.

Since the matching function (3.12) is symmetric in b and s, we can also rewrite it as

$$m(s, b) = s \cdot \left[1 + \theta^{-\sigma}\right]^{-1/\sigma},$$

When tightness is very large, $\theta^{-\sigma}$ is negligible compared to 1, so the matching function approximates the number of sellers: $m(s, b) \approx s$. This means that when the number of sellers is small, the number of sellers gives the number of trades, irrespective of the number of buyers.

Finally, the selling and buying probabilities given by the CES matching function and its matching elasticity are illustrated in figure 3.2.

3.7. Cobb-Douglas matching function

Maybe the most popular of all matching functions is the Cobb-Douglas matching function. It is appealing because it is even easier to manipulate than the CES matching function, and it describes the US labor market well (Blanchard and Diamond 1989; Petrongolo and Pissarides 2001). It is possible to obtain a Cobb-Douglas matching function from the queuing process proposed by Stevens (2007), as long as matching costs are linear. The main drawback of the Cobb-Douglas matching function is that it does not guarantee that $m(s, b) \leq \min(s, b)$, so it must be truncated in discrete-time models to avoid trading probabilities greater than 1.

3.7.1. Expression

The Cobb-Douglas matching function takes the following form:

$$m(s, b) = \mu \cdot s^{\sigma} \cdot b^{1-\sigma}. \tag{3.14}$$

The parameter $\mu > 0$ governs the efficacy of the matching process. The parameter $\sigma \in (0, 1)$ governs the shape of the matching function.

3.7.2. Properties

We easily check that the Cobb-Douglas function satisfies the assumptions listed in section 3.3.

The Cobb-Douglas matching function is clearly smooth, strictly positive, strictly increasing in each argument, and zero at the limit without buyers or sellers. It also clearly exhibits constant returns to scale:

$$m(zs, zb) = \mu(zs)^{\sigma}(zb)^{1-\sigma} = zm(s, b).$$

And it is clearly strictly concave in the number of sellers and in the number of buyers.

A reason why the Cobb-Douglas matching function is particularly convenient is that its matching elasticity is constant: $\eta = \sigma$.

Relatedly, the trading probabilities that it produces are easy to deal with. We obtain them directly using (3.4) and (3.5):

$$f(\theta) = m(1, \theta) = \mu\theta^{1-\sigma}, \quad q(\theta) = m\left(\frac{1}{\theta}, 1\right) = \mu\theta^{-\sigma}. \tag{3.15}$$

In particular, the trading probabilities have constant elasticity with respect to tightness: $\epsilon^{f}_{\theta} = 1 - \sigma$ and $\epsilon^{q}_{\theta} = -\sigma$.

The Cobb-Douglas matching function has one main limitation: it is not always less than the minimum of its two arguments. In a continuous-time model, where the matching function describes the flow of trades at any point in time, this is not an issue. But in a discrete-time model, where the matching function describes the number of trades in a period, this is an issue because it might lead to trading probabilities that are greater than 1. To avoid trading probabilities that are greater than 1, the matching function must be truncated by imposing $m(s, b) = \min\left(\mu \cdot s^{\sigma} \cdot b^{1-\sigma}, s, b\right)$.[3] The truncation in turn creates difficulties in theoretical and numerical work (den Haan, Ramey, and Watson 2000). Hence, when the truncation is binding, it is often better to use another matching function.

The selling and buying probabilities given by the Cobb-Douglas matching function and its matching elasticity are illustrated in figure 3.3.

3.8. Empirical properties of the matching function

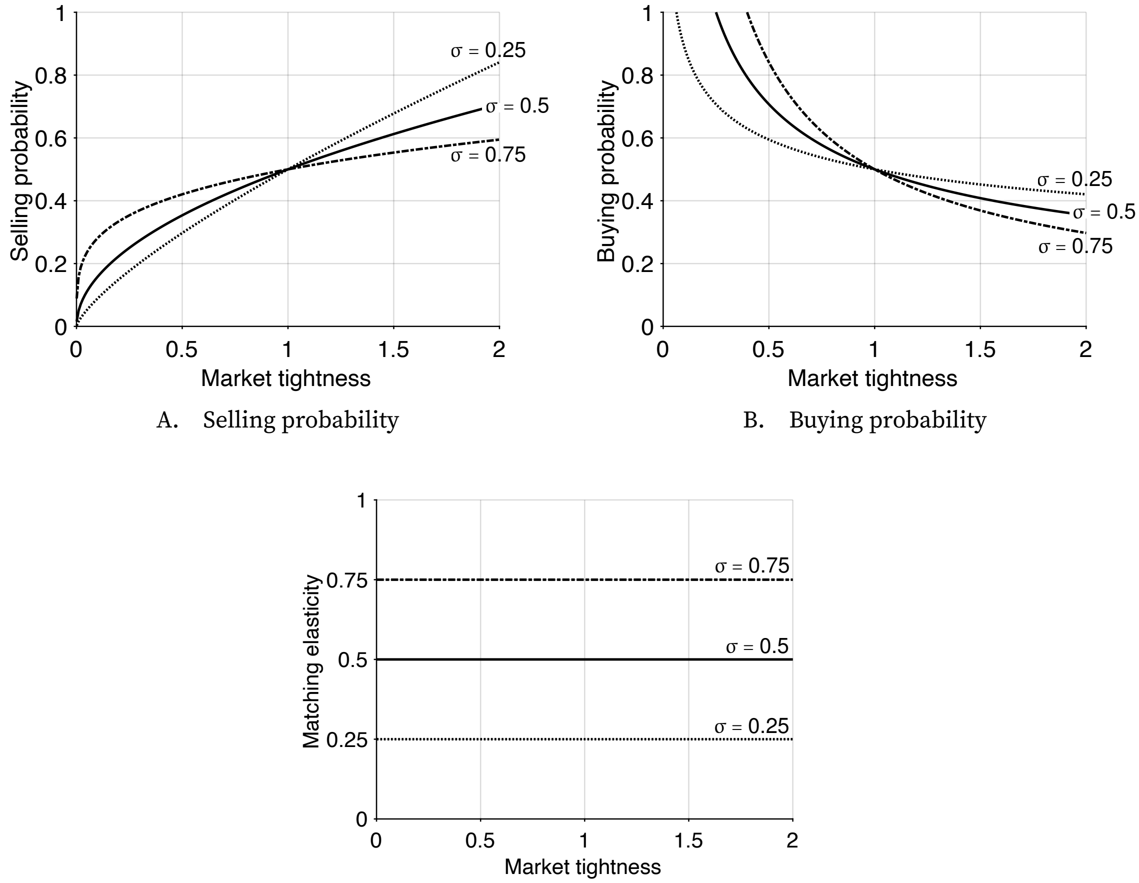
To conclude this chapter, we briefly review the empirical properties of the matching function to convince ourselves that our theoretical assumptions give us a matching function that describes the real world well. When matching functions were first developed, researchers examined the data to ensure that they built functions that were realistic (Blanchard and Diamond 1989; Petrongolo and Pissarides 2001). Here we review modern US data with the same goal.

Data on sellers and buyers are most readily available in the labor market, where we have good counts of who sells labor (job seekers) and who buys labor (vacant jobs). Thus we focus on the matching function for the labor market.

A central assumption is that the matching function has constant returns to scale. The main implication is that the market tightness determines the rates at which sellers and buyers trade. In the labor market, this means that labor market tightness determines the rate at which job seekers find jobs and the rate at which vacant jobs are filled. We only have data to compute the job-finding rate over a long period of time, so we concentrate on the link between job-finding rate and labor market tightness here. The data that we use start in 1948, which is when the BLS started to collect the data required to compute the job-finding rate (CPS data). Just like in figure 2.8, we stop the analysis in 2019 to concentrate on the 1948–2019 period, which is when the Beveridge curve and matching function are fairly stable.

For reference, we plot the US labor market tightness between 1948 and 2019. Labor market tightness is computed as the number of job vacancies per job seeker, or equiva-

[3] The expressions (3.15) show that $f(\theta)$ is larger than 1 when θ is high enough, and $q(\theta)$ is larger than 1 when θ is low enough. This is why the Cobb-Douglas matching function must be truncated in discrete-time models.



A. Selling probability

B. Buying probability

C. Matching elasticity

FIGURE 3.3. Properties of the Cobb-Douglas matching function

The Cobb-Douglas matching functions are given by (3.14) with $\mu = 0.5$ and $\sigma = 0.25$, $\sigma = 0.5$, $\sigma = 0.75$. The selling and buying probabilities are computed from the matching functions with (3.4) and (3.5). The matching elasticity is constant: $\eta = \sigma$.

lently, the ratio of vacancy rate to unemployment rate.[4] We see that labor market tightness is sharply procyclical. It averages 0.65 between 1948 and 2019. Over that period, tightness reached its lowest level, 0.16, during the Great Recession (2009:Q3) and its highest level, 1.60, during the Korean War (1953:Q1).

We start with the rate at which job seekers find jobs, and correlate this rate with labor market tightness. To compute the job-finding rate, we follow Shimer (2012). In month t, $u(t)$ workers are looking for jobs. Some of them find a job in the month, and some do not. Those who do not find a job remain unemployed, so $u(t) - u(t+1)$ seems to indicate the number of workers who have been able to find a job. There is a complication,

[4]In practice some vacancies are filled not by unemployed workers but by employed workers moving between jobs. The present empirical approach is valid as long as the number of vacancies captures well the demand for unemployed labor.

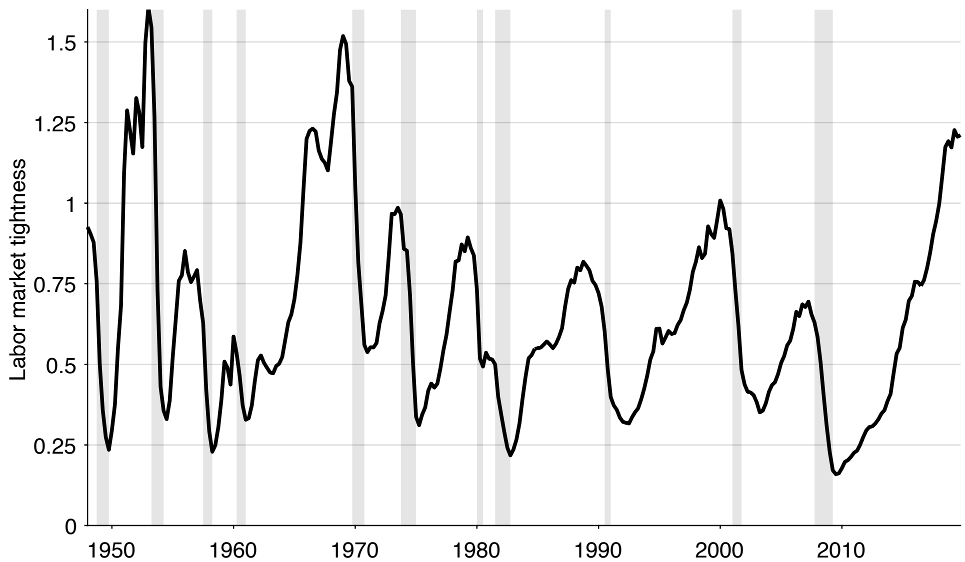


FIGURE 3.4. Labor market tightness in the United States, 1948–2019

Labor market tightness is the ratio of vacancy rate to unemployment rate. The unemployment rate comes from figure 2.1 while the vacancy rate comes from figure 2.5. Shaded areas indicate recessions dated by the NBER (2023).

however. Some workers who were employed lose their jobs in month t and join the pool of unemployed workers. We need to account for those when we compute the number of job seekers who found a job. We denote by $u^s(t+1)$ the number of workers who were previously employed and who have joined unemployment between months t and $t+1$. The number of workers who have found a job within month t is $u(t) - [u(t+1) - u^s(t+1)]$.[5] Dividing this number of job finders by the number of job seekers in month t, we obtain the probability of finding a job in month t:

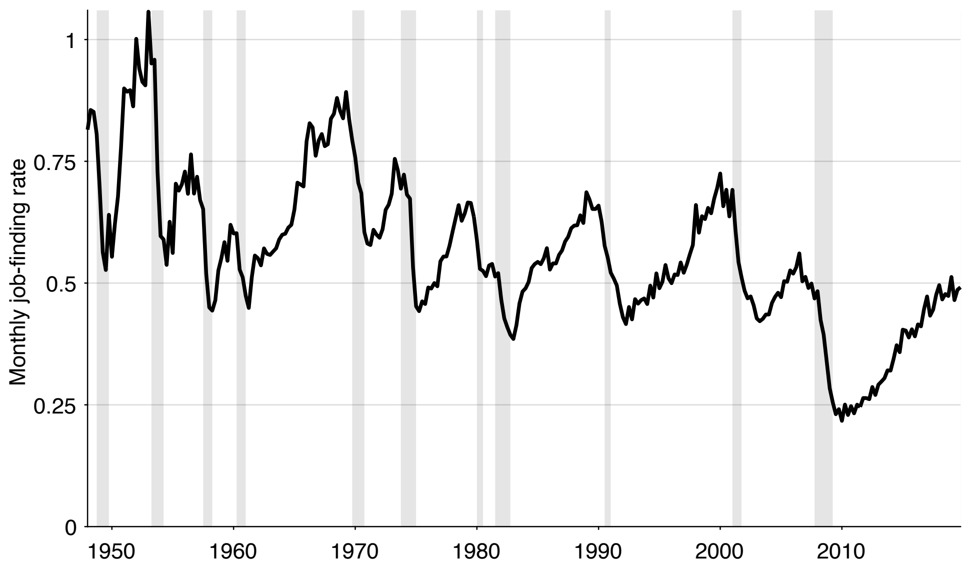
$$F(t) = 1 - \frac{u(t+1) - u^s(t+1)}{u(t)}.$$

To calculate $F(t)$, we measure $u(t)$ as the number of unemployed persons in month t computed by the BLS (2025b), and $u^s(t)$ as the number of persons who have been unemployed for less than 5 weeks in month t computed by the BLS (2025a).[6]

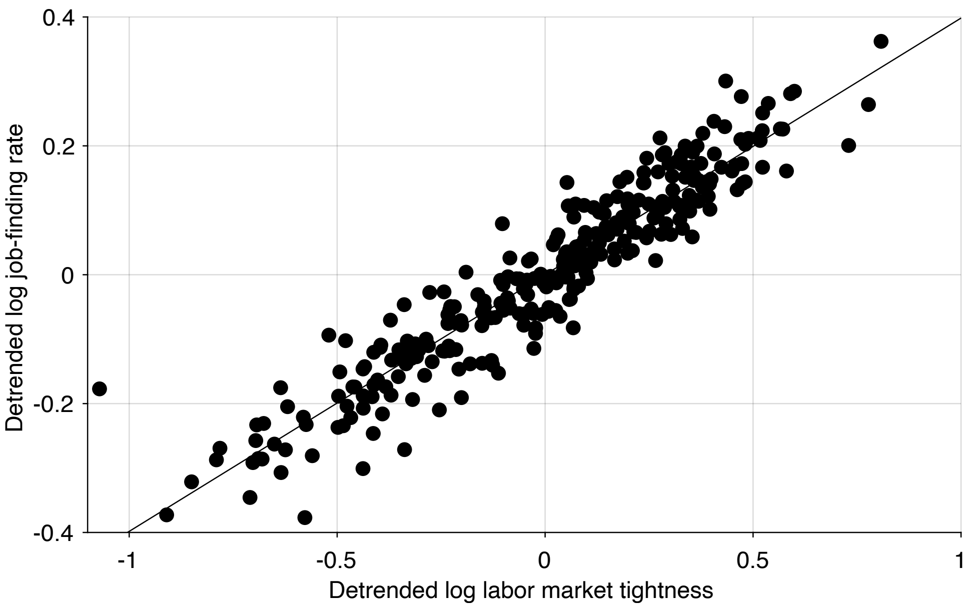
Assuming that unemployed workers find a job according to a Poisson process with monthly arrival rate $f(t)$, the probability that they have found a job within one month is

[5]This calculation assumes that workers only transition between employment and unemployment. In reality, some workers move in and out of the labor force. But the approach is valid as long as the numbers of job seekers who exit and enter the labor force are similar.

[6]Following Shimer (2012, appendix A), we multiply the series for $u^s(t)$ by 1.1 after January 1994 to correct for a change in the design of the CPS.



A. Monthly job-finding rate in the United States, 1948–2019



B. Detrended logarithm of tightness and job-finding rate, 1948–2019

FIGURE 3.5. The US job-finding rate is an isoelastic function of labor market tightness

The monthly job-finding rate is computed from (3.16). Labor market tightness comes from figure 3.4. In panel A, shaded areas indicate recessions dated by the NBER (2023). In panel B, the series are detrended by applying a HP filter with smoothing parameter of 10,000.

$F(t) = 1 - \exp(-f(t))$ (result A.4). Hence, we infer the job-finding rate from the job-finding probability:

$$f(t) = -\ln(1 - F(t)). \tag{3.16}$$

We thus obtain the monthly job-finding rate in the United States (figure 3.5A). Between 1948 and 2019, the job-finding rate is highly procyclical and averages 56% per month. This means that on average, a US job seeker takes $1/0.56 = 1.8$ months to find a job.

Our objective is to assess whether labor market tightness determines the job-finding rate. To uncover the relationship between tightness and job-finding rate, we plot the logarithm of the job-finding rate against the logarithm of tightness. To remove slow changes in the matching function, we detrend both series using an HP filter with smoothing parameter 10,000. We find that log job-finding rate is largely determined by log labor market tightness: the least-squares regression gives $R^2 = 0.89$, with a coefficient of 0.40 (figure 3.5B). This result indicates that the job-finding rate is well described as an isoelastic function of tightness, with an elasticity of 0.40.

Overall, we find that the job-finding rate is a stable function of tightness, which supports the assumption that the matching function has constant returns to scale. In addition, we have seen that the elasticity of the job-finding rate with respect to tightness appears constant, which suggests that the matching elasticity is constant. This finding suggests that a Cobb-Douglas matching function, which has a constant matching elasticity, describes the US labor market better than the urn-ball or CES matching functions, which have matching elasticities that respond noticeably to tightness (figures 3.1C and 3.2C). Furthermore, the matching elasticity that appears in the regression is $\eta = 0.60$, using (3.8).

The empirical findings in this section are in line with findings in the literature on the matching function. Early aggregate studies of the US labor market find that the matching function involves the stock of unemployed workers and job vacancies with constant returns to scale (Petrongolo and Pissarides 2001). The studies converge on a Cobb-Douglas form with a matching elasticity between 0.5 and 0.7. More recent studies obtain comparable results. Shimer (2005, table 2) estimates the matching elasticity at 0.72; Rogerson and Shimer (2011, p. 638) obtain an estimate of 0.58; and Borowczyk-Martins, Jolivet, and Postel-Vinay (2013, table 1) report a lower estimate of 0.30.

3.9. Summary

In this chapter we introduce the matching function as a key tool for modeling slackish markets, where trade is not instantaneous and requires time and effort from both buyers and sellers. Unlike Walrasian markets where trades are seamless, real-world markets are characterized by complexities that make it difficult for buyers and sellers to find each other. The matching function is an aggregate tool, similar to a production function, that captures the outcome of this complex trading process without modeling the micro-level details explicitly.

The chapter presents three specific types of matching functions: urn-ball, CES, and Cobb-Douglas, which each have their advantages and drawbacks, which we discussed. For instance, the CES function is useful in theoretical work, while the Cobb-Douglas function is especially parsimonious.

Finally, the chapter reviews empirical evidence on matching from the US labor market, which supports the use of a matching function with constant returns to scale. Furthermore, this evidence suggests a Cobb-Douglas matching function is a good approximation for the US labor market, because the matching elasticity appears roughly constant.

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